THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A tree casts a shadow of 10 m when a 2 m stick casts a 1 m shadow at the same time. The height of the tree is: 1

(a) 10 m **(b)** 12 m **(c)** 5 m **(d)** 20 m

2. In the northern hemisphere a waxing crescent Moon is seen in the: 1

(a) morning sky only **(b)** sky at midnight
(c) evening sky, in the west **(d)** sky at noon

3. Why does vinegar taste sour? 1

(a) it contains sodium **(b)** it contains sugar
(c) it contains acetic acid **(d)** it contains salt

4. A bulb uses a battery of two cells. If a third cell is added in series, the bulb: 1

(a) goes off **(b)** becomes dimmer **(c)** becomes brighter **(d)** becomes cold

5. Why does an aluminium vessel not corrode easily? 1

(a) a protective oxide layer forms **(b)** it is a non-metal
(c) it is sonorous **(d)** it is brittle

6. Which statement about crystallisation is true? 1

(a) crystals form from a gas always
(b) crystals form by sieving
(c) crystals form by burning
(d) crystals form when a hot saturated solution cools

7. The thyroid gland is in the: 1

(a) brain **(b)** abdomen **(c)** neck **(d)** chest

8. Which of these clothes would be best on a very hot, sunny day? 1

(a) black polyester **(b)** leather
(c) white loose cotton **(d)** woollen

9. In a distance-time graph, the slope gives: 1

(a) speed **(b)** distance only **(c)** time **(d)** acceleration

10. The heart of a human has: 1

(a) three chambers **(b)** two chambers **(c)** five chambers **(d)** four chambers

11. Cuscuta has no chlorophyll. It survives by: 1

(a) taking food from its host **(b)** making food
(c) eating insects **(d)** absorbing from dead matter

12. In a pinhole camera, if the object distance is doubled, the image size becomes: 1

(a) double **(b)** same **(c)** quarter **(d)** half

13. During a total solar eclipse, the observer is in the Moon's: 1

(a) penumbra **(b)** sunlight **(c)** umbra **(d)** shadow of Earth

14. A solution that turns red litmus blue and turmeric red is: 1

(a) neutral **(b)** sugary **(c)** basic **(d)** acidic

15. Two cells of 1.5 V are connected with opposite polarity in series. The net voltage is about: 1

(a) 4.5 V **(b)** 0 V **(c)** 3 V **(d)** 1.5 V

16. A substance is dull, brittle and does not conduct electricity. It is probably a: 1

(a) non-metal **(b)** magnet **(c)** alloy **(d)** metal

17. Which one of these pairs contains only physical changes? 1

(a) cooking, curdling **(b)** burning wood, rusting
(c) freezing of water, evaporation of water **(d)** digestion, photosynthesis

18. A boy's voice breaks at 14. It shows that his: 1

(a) larynx is growing **(b)** teeth are growing
(c) heart is growing **(d)** lungs are growing

19. Heat reaches us from a bonfire at a distance mainly by: 1

(a) radiation **(b)** evaporation **(c)** convection **(d)** conduction

20. A runner covers 1.5 km in 10 min. His speed in km/h is: 1

(a) 1.5 km/h **(b)** 15 km/h **(c)** 9 km/h **(d)** 0.15 km/h

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. A car travels 150 km in 2.5 h. Find its speed in km/h and m/s. 2

22. Why does the number of breaths per minute rise during exercise? 2

23. Why are potted plants kept near a window grown healthier than those kept in the dark? 2

24. The angle of incidence of a ray on a plane mirror is 35°. Find the angle of reflection and the angle between incident and reflected rays. 2

25. Why do days and nights have unequal lengths in different seasons? 2

26. Write a neutralisation reaction of sulphuric acid with sodium hydroxide. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Design an experiment to show that CO2 is needed for photosynthesis. 3

28. A ray of light makes an angle of 70° with the surface of a plane mirror. Find the angle of incidence, reflection and the angle between incident and reflected rays. 3

29. Explain with a diagram in words how a total solar eclipse is formed. 3

30. Why are some stomach pills chewed and others dissolved? 3

31. Describe how an MCB works in simple terms. 3

OR

Explain why a tarnished copper vessel can be cleaned with lemon or tamarind. 3

32. Why is photosynthesis a chemical change? 3

OR

What are the factors that lead to tobacco use among teens and how to prevent it? 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. A ray of light strikes a plane mirror at 60° to the surface. Draw and describe the path of the reflected ray and find all angles. 5

34. Draw (describe) the positions of the Earth in its orbit at the two solstices and the two equinoxes and write the seasons in both hemispheres. 5

35. A student mixes 10 mL of vinegar with baking soda. Describe the observation and explain the reaction in words. 5

36. Describe how an electrician stays safe while repairing a socket. 5

OR

Describe a safe procedure to burn magnesium and test the product. 5

37. Discuss the harmful and useful chemical changes with three examples each. 5

OR

Explain the emotional changes in adolescence and give guidance to teenagers for managing them. 5

38. Describe the construction of a solar water heater and explain the heat transfer in it. 5

OR

Explain why a pendulum clock stops after a while and how it is kept going. 5

END OF QUESTION PAPER — Total: 80 Marks

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